Data-driven method application for PDE identification and dynamic prediction using PDEBench data

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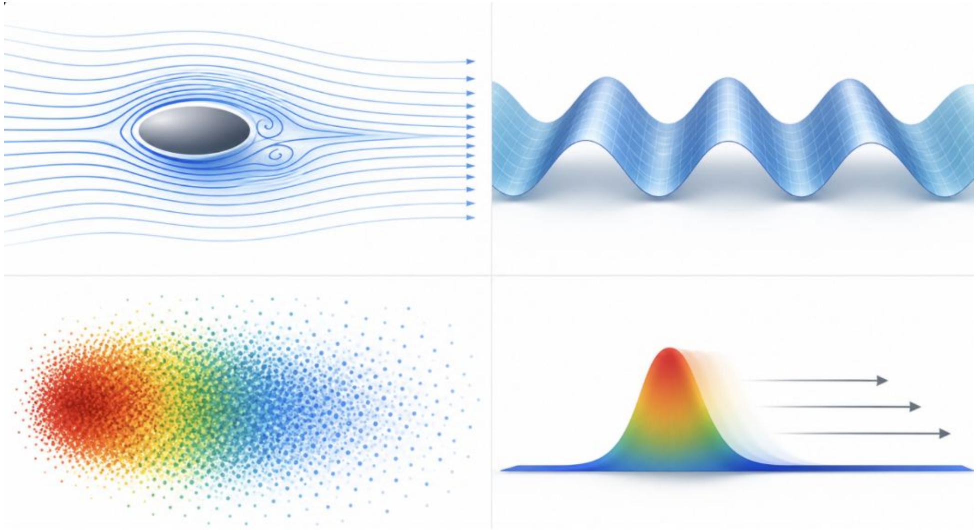
Introduction

- Many physical processes are described by PDEs: fluid flow, waves, diffusion, transport.
- Classical numerical solvers can be computationally expensive.
- Data-driven methods can learn dynamics directly from simulation data.
- The key question: **can we identify the underlying PDE and predict future states using only data?**

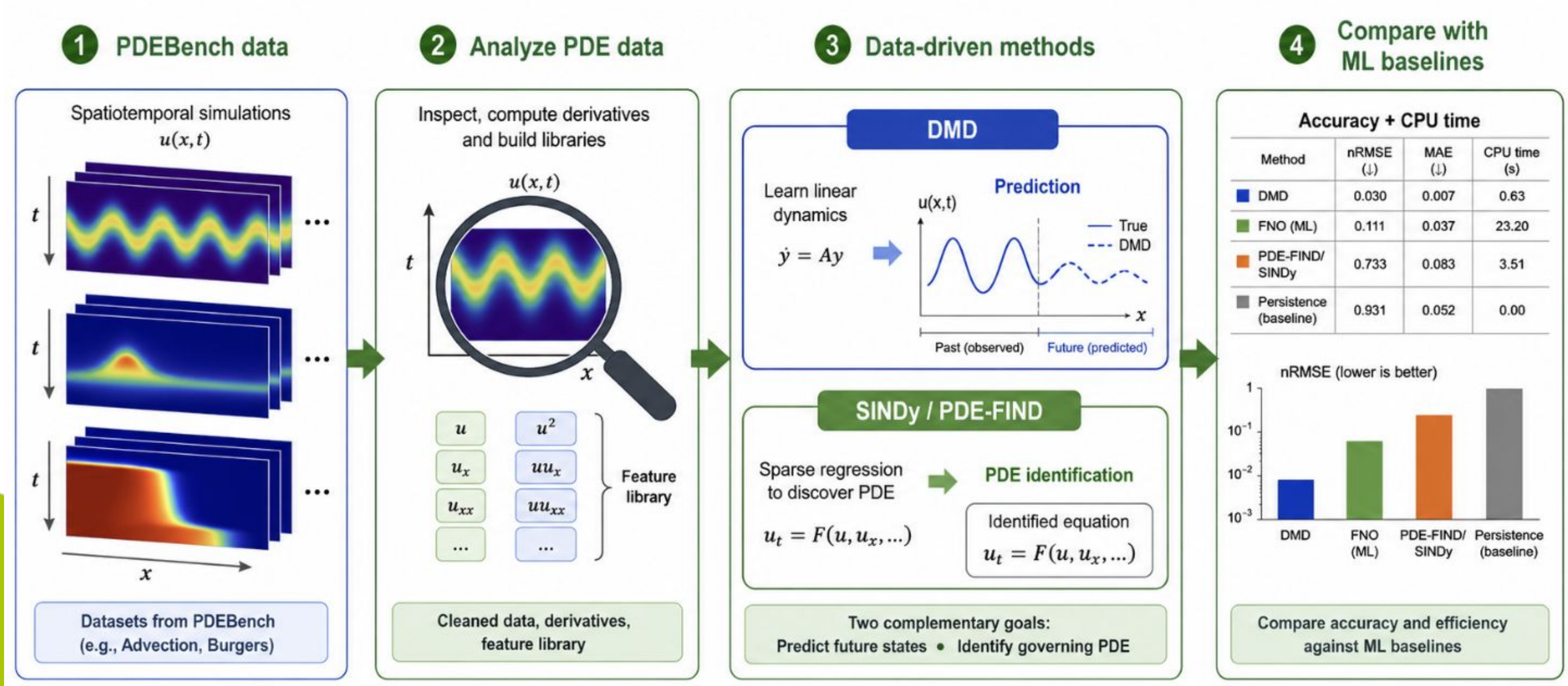


Introduction

The goal: apply data-driven methods to PDEBench datasets in order to identify the underlying PDE dynamics and predict future system states:

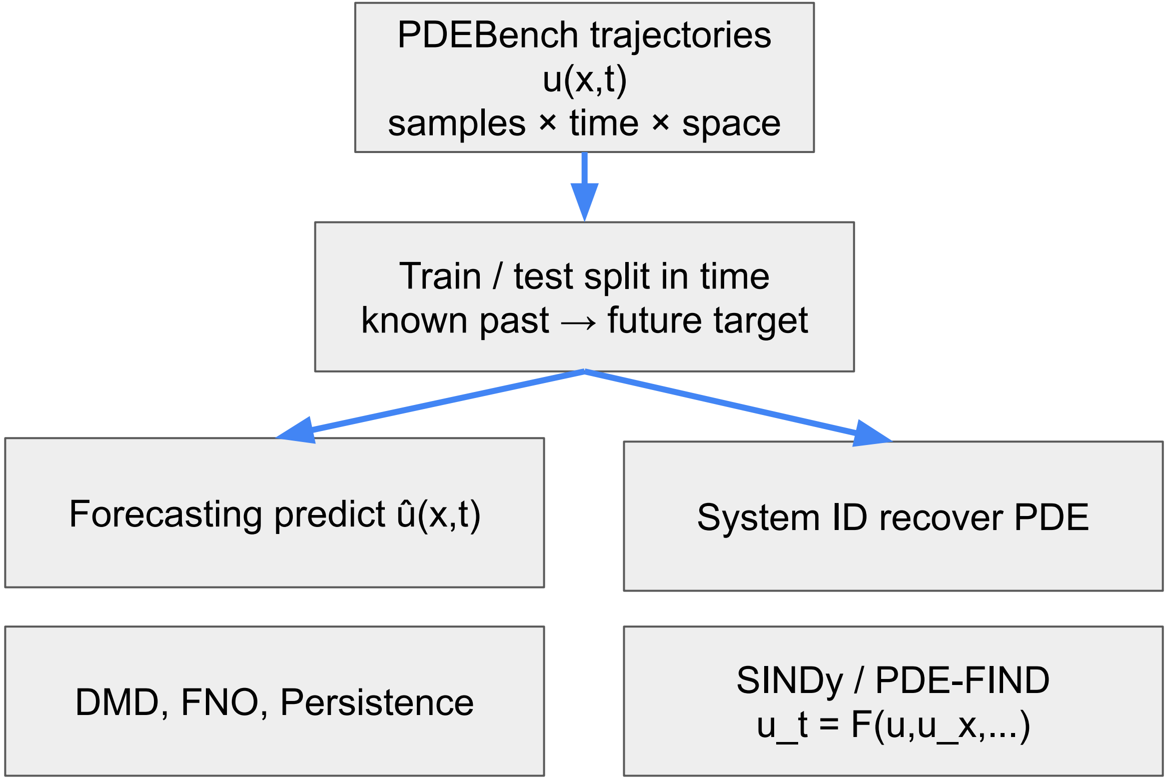
- Analyze PDE simulation data.
- Apply DMD and SINDy-based methods for dynamics identification and reconstruction.
- Compare data-driven results with machine learning baselines using accuracy and computational efficiency metrics.

Introduction



Goal: Use data-driven methods on PDEBench datasets to identify the underlying PDE dynamics and predict future system states.

What problem are we solving?



PDEBench, Advection and Burgers

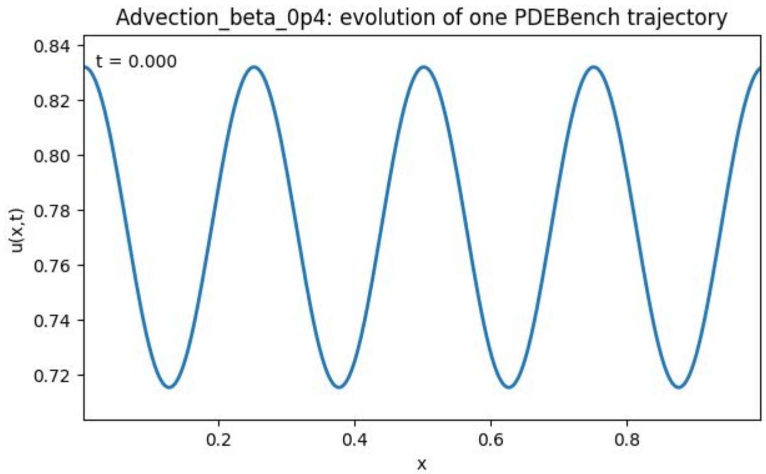
What is u(x,t)?
u(x,t) is a field value depending on:
x — spatial coordinate
t — time
One trajectory = full evolution of u over space and time

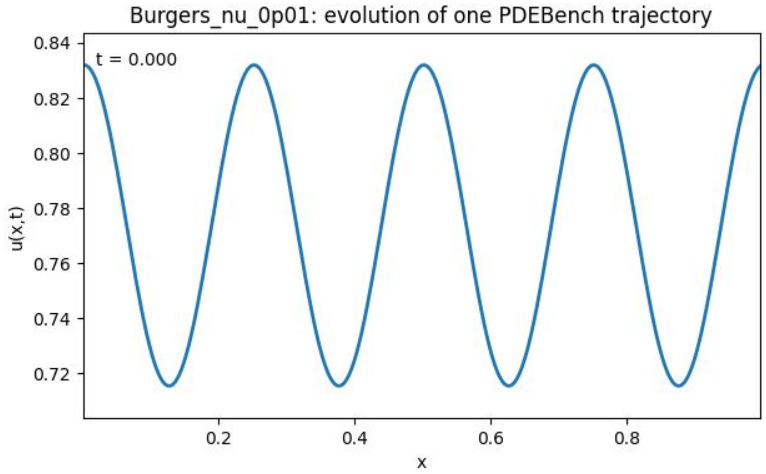
Advection: linear transport — the profile moves along x with almost **preserved** shape

$$u_t = -0.4u_x$$

Burgers: nonlinear transport + diffusion — the profile deforms and smooths over time

$$u_t = -uu_x + 0.01u_{xx}$$





General project goals

The expected project work can cover the following tasks:

- Perform data parsing, processing and visualization
- Get understanding about the data and **repeat** the machine learning benchmarks on pre-training machine learning models
- Perform the analysis of the spatiotemporal data based on DMD approach or its modification to identify the dynamics in this case
- Apply SINDy or its modification to see if original PDE can be reconstructed
- **Analyse DMD and SINDy reconstruction results**
- **Compare the results** of pre-trained machine learning and data-driven methods in terms of accuracy (MAE and/or MAPE) and computational efficiency (CPU time for training and prediction)
- ***Conclude about the key factors and approach for improving the system identification and reconstruction***

DMD and SINDy reconstruction results

1D Advection Equation Identification

Expected Governing Equation:

$$u_t = -0.4u_x$$

Identified PDE:

$$u_t = -0.400047u_x$$

We got highly accurate recovery of the wave propagation speed. All other candidate terms in the library (such as u, u^2, uxx) were successfully regularized and dropped to near-zero coefficients.

1D Burgers' Equation Identification

Expected Governing Equation: $u_t = -1.0uu_x + 0.01u_{xx}$

Identified PDE: $u_t = -0.928976uu_x + 0.003553u_{xx} + 0.109495u^3 - 0.079721u + 0.047363u^2$

The non-linear advection term is recovered with high accuracy (-0.93 vs. -1).

The viscosity coefficient is underestimated (0.0036 vs. 0.01) due to derivative approximation noise.

Discretization errors also introduced minor spurious terms (e.g. u, u^2 , u^3).

Comparing the results

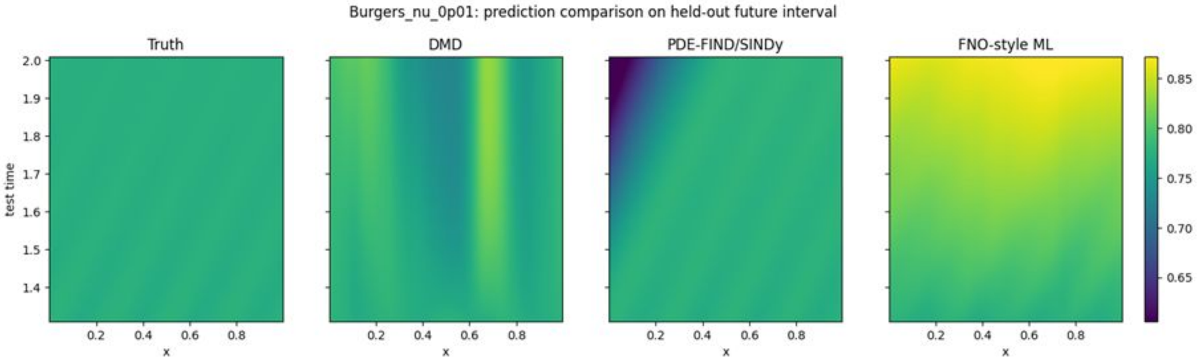


Figure 1. Burgers forecast comparison on the held-out interval: ground truth, DMD, PDE-FIND/SINDy, and compact FNO-style baseline.

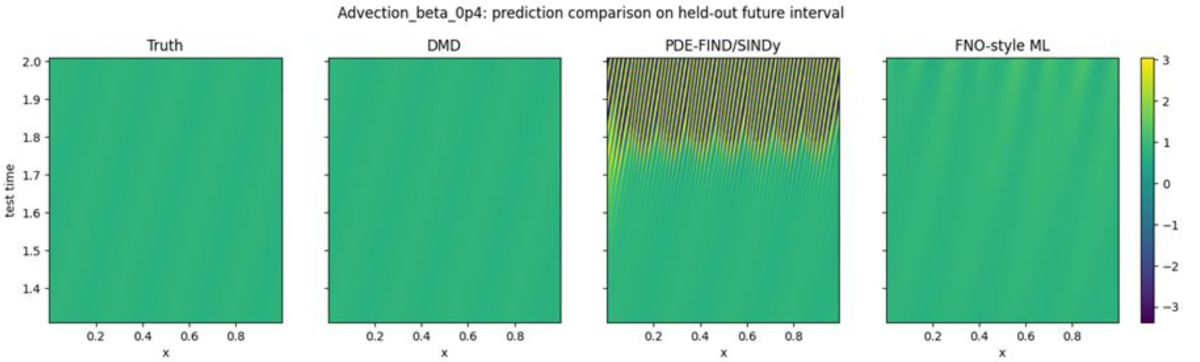


Figure 2. Advection forecast comparison on the held-out interval. DMD follows the linear transport dynamics most closely.

Comparing the results

	dataset	pde	method	train_seconds	prediction_seconds	total_seconds	RMSE	nRMSE	MAE
0	Advection_beta_0p4	Advection	DMD	0.538646	0.086833	0.625479	0.021106	0.030029	0.006632
1	Advection_beta_0p4	Advection	FNO-style ML	20.422785	0.220627	20.643412	0.573843	0.816466	0.261692
2	Advection_beta_0p4	Advection	Persistence	0.000000	0.000000	0.000000	0.654168	0.930754	0.451053
3	Advection_beta_0p4	Advection	PDE-FIND/SINDy	1.279502	2.419066	3.698568	1.653924	2.353211	1.020656
4	Burgers_nu_0p01	Burgers	FNO-style ML	23.030923	0.165851	23.196775	0.059344	0.111087	0.037098
5	Burgers_nu_0p01	Burgers	DMD	0.536496	0.091803	0.628299	0.073658	0.137882	0.045793
6	Burgers_nu_0p01	Burgers	Persistence	0.000000	0.000000	0.000000	0.090953	0.170256	0.051812
7	Burgers_nu_0p01	Burgers	PDE-FIND/SINDy	1.460026	2.051761	3.511787	0.391461	0.732783	0.082924

Conclusions

1 **DMD Superiority on Linear Transport:** For the Advection dataset, DMD achieves **excellent accuracy** (nRMSE=0.03) with a total execution time of under seconds.

2 **FNO Accuracy on Non-linear Shocks:** Our compact FNO-style neural network yields **the lowest error on the Burgers' dataset** (nRMSE=0.111), capturing the non-linear shock front deformation.

However, it requires a **higher training budget** (18,5 seconds) and **struggles on purely hyperbolic advection** due to limited training epochs (20 epochs).

3 **Rollout Instability in SINDy:** Despite **accurate equation identification**, PDE-FIND/SINDy **suffers from numerical error propagation** (blow-up) during explicit Euler integration, leading to poor long-term forecasting metrics.

LIterature list

- https://arxiv.org/abs/2210.07182
- https://darus.uni-stuttgart.de/file.xhtml?fileId=281363&version=8.0
- https://darus.uni-stuttgart.de/file.xhtml?fileId=255674&version=8.0
- https://github.com/pdebench/PDEBench
- https://github.com/pdebench/PDEBench/blob/main/pdebench/data_gen/notebooks/Analysis.ipynb

Thank you for attention!